

Urban Air Quality & Public Health Impact Analyzer

Problem Statement Document

Data Analysis Project · Python · MySQL · Power BI

1. Background

Air pollution has emerged as one of the most critical environmental and public health challenges in urban India. With rapid industrialization, increasing vehicular traffic, and urban expansion, major Indian cities face dangerously high levels of particulate matter and toxic gases. According to the World Health Organization (WHO), 9 out of 10 people worldwide breathe polluted air, and India accounts for a significant portion of global air pollution-related deaths annually.

Despite the availability of air quality monitoring infrastructure through the Central Pollution Control Board (CPCB), actionable insights connecting pollution levels to public health outcomes remain underutilized. There is a critical need for data-driven analysis that translates raw monitoring data into meaningful insights for policymakers, healthcare administrators, and urban planners.

2. Problem Statement

"How does air pollution across major Indian cities impact public health, and which cities, seasons, and pollutants pose the greatest risk to citizens — and what data-driven interventions can reduce this impact?"

3. Project Objectives

- **Analyze air quality patterns** across 30 major Indian cities over 2022–2023 using AQI and pollutant data.
- **Identify seasonal pollution trends** and determine which seasons pose the greatest public health risk.
- **Quantify the health impact** of air pollution by correlating AQI and PM2.5 levels with hospital admissions and respiratory cases.
- **Compare pollution by zone type** to understand how Industrial, Urban, Coastal, and IT Hub zones differ in pollution levels.

- **Engineer meaningful features** such as AQI Category and Health Risk Score for deeper analysis.
- **Build an interactive dashboard** in Power BI for visual exploration of pollution and health data.
- **Provide actionable recommendations** for urban planners, health authorities, and policymakers.

4. Key Business Questions

This project will answer the following 8 business questions through SQL analysis and Power BI visualizations:

#	Business Question	Tool
1	What is the overall air quality status across all cities?	MySQL + KPI Card
2	Which are the top 5 most polluted cities by average AQI?	MySQL + Bar Chart
3	Which season has the worst air quality and health impact?	MySQL + Heatmap
4	Which pollutant has the highest average concentration?	MySQL Query
5	How does AQI category relate to hospital admissions?	MySQL + Scatter Plot
6	Which zone type (Industrial/Urban/Coastal) is most polluted?	MySQL + Column Chart
7	Which states record the most Hazardous air quality days?	MySQL Query
8	Is there a measurable difference in AQI on weekdays vs weekends?	MySQL Query

5. Dataset Overview

Attribute	Detail
File Name	urban_air_quality_raw.csv
Total Rows	3,000 (raw) → 2,732 after cleaning
Total Columns	33
Cities	30 major Indian cities across 18 states
Date Range	January 2022 – December 2023
Key Columns	AQI, PM2.5, PM10, NO2, SO2, CO, O3, NH3
Health Columns	Hospital Admissions, Respiratory Cases, Cardiovascular Cases
Derived Columns	AQI Category, Health Risk Score (added during cleaning)
Data Issues	~4% missing AQI, ~3% missing PM2.5, AQI outliers >500, negative wind speed

6. Project Scope

In Scope:

- 30 major Indian cities across multiple states and zone types
- Daily air quality readings from January 2022 to December 2023
- 7 pollutants: PM2.5, PM10, NO2, SO2, CO, O3, NH3
- Weather variables: temperature, humidity, wind speed, rainfall
- Health outcomes: hospital admissions, respiratory and cardiovascular cases
- Full pipeline: Python cleaning → MySQL analysis → Power BI dashboard

Out of Scope:

- Real-time live data streaming
- Predictive machine learning models (future enhancement)
- Individual patient-level health records
- Economic cost modeling of health impacts

7. Tools & Methodology

Step	Tool	Task
1	Python + Jupyter	Data cleaning, feature engineering, EDA, visualization
2	MySQL Workbench	Data import, SQL queries, business question analysis
3	Power BI Desktop	Interactive dashboard, DAX measures, slicers
4	Word / PDF	Project report and documentation
5	GitHub	Version control and portfolio hosting

8. Expected Outcomes

- A fully cleaned dataset ready for analysis with no missing or invalid values
- 8 SQL queries answering key business questions with result tables
- An interactive Power BI dashboard with 6 charts, 5 KPI cards, and 4 slicers
- Identification of the most polluted cities, worst seasons, and highest-risk pollutants
- Quantified relationship between PM2.5 levels and hospital admissions
- 6 data-driven business recommendations for health authorities and urban planners
- A complete project report documenting methodology, findings, and insights
- GitHub repository with organized folder structure for public portfolio

9. Success Criteria

Criteria	Target
Data Cleaning	Zero missing values, zero duplicates in final dataset
SQL Coverage	All 8 business questions answered with query results
Dashboard Completeness	6 charts + 5 KPI cards + 4 slicers fully functional
Insight Quality	At least 5 key findings documented with supporting data
Recommendations	Minimum 6 actionable business recommendations
Documentation	Complete project report with screenshots
GitHub Upload	Organized repo with README, data, notebooks, SQL, dashboard